

8ai	Equatorial plane. West to east. (i.e. follow Earth's rotation about its axis)	B1 B1
8aii	Advantages: (either one) 1. As the satellites <u>remain stationary above the same point on Earth</u> , they are ideal for use as communication satellites as they <u>require no tracking</u> to receive signals from Earth. 2. As geostationary satellites are <u>positioned at such a high altitude</u> , they can <u>capture almost the full disk Earth image</u> . Disadvantage: (either one) 1. As geostationary satellites are positioned at a high altitude, the <u>spatial resolution of their images tends to be not as good</u> as satellites which are much closer to Earth. 2. Since the geostationary satellites are positioned above the equator, they <u>cannot see the north or south poles and are of limited use for latitudes greater than 60-70 degrees north or south</u> .	B1 B1
8bi	For circular orbit, gravitational force of attraction between the stars provides the centripetal force for each other, which are directed towards the centre of the circular orbit.	B1 B1
8bii	Gravitational force provides centripetal force, $\frac{Gmm}{R^2} = m\left(\frac{R}{2}\right)\left(\frac{4\pi^2}{T^2}\right)$ $\therefore m = \frac{2\pi^2 R^3}{T^2 G} = \frac{2\pi^2 (3.6 \times 10^{10})^3}{(20.5 \times 86400)^2 6.67 \times 10^{-11}}$ $= 4.4 \times 10^{30} \text{ kg}$ Able to calculate angular velocity [1] Able to calculate centripetal force or gravitational force[1] correct answer [1]	
8ci	$V_s = -\frac{GM_s}{R_s}$ $= -\frac{(6.67 \times 10^{-11})(5.68 \times 10^{26})}{5.82 \times 10^7}$ $= -651 \times 10^6 \text{ J kg}^{-1}$ $= -651 \text{ MJ kg}^{-1}$	M1 A1
8cii	651 MJ of work is required to move a 1 kg mass from the surface of Saturn to a point at infinity OR 651 MJ is done by the external force to bring a 1 kg mass from infinity to the surface of Saturn. (Accept other alternative answers, e.g. minimum KE at launch from surface to escape totally etc)	1

8di	The resultant gravitational pull is towards Saturn (since Saturn has a larger mass than Titan)	1
	Probe is moving in opposite direction to net gravitational pull, so it is gaining GPE.	1
8dii	No resultant gravitational force $\Rightarrow$ no net g $\frac{Gm_s}{d_s^2} = \frac{Gm_T}{d_T^2}$ $\frac{d_T^2}{d_s^2} = \frac{m_T}{m_s}$ $\therefore \frac{d_T}{d_s} = \sqrt{\frac{m_T}{m_s}} = \sqrt{\frac{1.35 \times 10^{23}}{5.68 \times 10^{26}}} = 0.0154$ Hence the neutral point is at $\frac{0.0154}{1.0154}(1.22 \times 10^6) = 18,500 \text{ km}$	M1 M1 A1
8ei	$K_T > m(V_E - V_T)$ (Accept $K_T > m V_E - V_T $ and $K_T > m(V_T - V_E)$)	1
8eii	$K_S + m(V_T - V_S)$ (Accept $K_S + m V_T - V_S $ and $K_S + m(V_S - V_T)$)	1